Start

Input Data

- Load profile (All India / IIT Kanpur)
- Solar PV generation (Kaggle)
- Electricity price (IEX / Mendeley)
- EV parameters (arrival, departure, SOC, battery, max power)
- EVCS / Port configuration (BEE)
- System parameters (grid/solar capacity, efficiency, etc.)

Data loaded successfully?

No → **Show error / missing file message** → Input Data

Yes

Data Preprocessing

- Convert to 24-hour horizon
- Create 15-min slots (T = 96)
- Format matrices & parameters

Processed Data

- Load profile (96 × n)
- Solar generation (96 × 1)
- Price (96 × 1)
- EV / EVCS data (structured)

Create MILP Decision Variables

- x(e,p,t) – EV-port assignment (binary)
- P(e,p,t) – Charging power (continuous)
- Ppv(e,p,t) – Solar power (continuous)
- Pg(e,p,t) – Grid power (continuous)
- SOC(e,t) – State of Charge (continuous)

Build Constraints

- EV arrival / departure
- One port per EV
- Port availability
- Station capacity
- EV max power
- Port power limit
- Grid capacity
- Solar capacity
- Power balance (P = Ppv + Pg)
- SOC update
- Required SOC at departure
- SOC bounds (0 – 100%)

Construct Objective Function

System Profit
- Charging revenue
- Grid energy cost
- Operating cost

Participant Profit
- Charging revenue
- Charging cost

Solar Utilization
- Maximize solar energy used

Weighted Objective (w1 = 40%, w2 = 40%, w3 = 20%)

$$\text{Maximize } Z = w1 \cdot P_{system} + w2 \cdot P_{participant} + w3 \cdot E_{solar}$$

(Convert to minimization: min (−Z))

Convert to MATLAB MILP Form

- $\mathbf{f^T x}$ (objective vector)
- **Ax ≤ b** (inequality constraints)
- **Aeqx = beq** – (equality constraints)
- **lb ≤ x ≤ ub** – (variable bounds)
- **intcon** – (integer variables)

MILP Structure

- f – objective coefficients
- A, b – inequality constraints
- Aeq, beq – equality constraints
- lb, ub – bounds
- intcon – integer indices

Solve using intlinprog

(MILP Solver)

Solver Process

- Find feasible solutions
- Check constraints
- Evaluate objective
- Select optimal solution

Optimal solution found?

No → **Adjust parameters / Reformulate** → Construct Objective Function

Yes

Extract Results (xsol)

- EV-port schedule
- Charging power (per slot)
- Solar / Grid allocation
- SOC profile

Validate Solution

- SOC requirement met?
- Power limits satisfied?
- Station capacity OK?
- Grid/Solar limits OK?
- Energy balance OK?

Generate Graphs & Analysis

- EV charging profile
- Solar vs Grid
- SOC profiles
- Station utilization
- Energy & economic results

End